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Suzhou High School - International Division

SZZX-ID Bridge Program Physics

Unit 3 Lesson 2: Circular motion

Class Outline

Title of Unit	Projectile and circular motions
Title of Lesson	Circular motion
Starter Activity	Find circular motions around us
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	Demonstration: Circular motion and centripetal force
	Lecture: The definition of the period, frequency, angular velocity, and linear speed
	Radium (rad) as a SI unit for angle
	Inquiry: how to describe a circular motion; design an experiment to determine g
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
Lesson Objectives	Give individual guidance to use calculator to calculate centripetal force and acceleration
	International Mindedness:
ATLs	By the end of the lesson, students will be able to:
	- understand how motion in a circle is described mathematically. - be able to describe examples of circular motion qualitatively and quantitatively. - be able to solve problems with angular characteristics. - Understand and apply the formula for centripetal force and acceleration - Know that many types of force can serve as centripetal force
IB Learner Profile	Identify unit ATLs and how they are being addressed in the lesson
	Communication
Teaching Activities	Social
	Link the lesson to the characteristics of the IB Learner attribute
Plenary Session	thinker
	Demonstration: observe and feel circular motion
	Practice using calculator: acceleration of a satellite
	Group discussion: design an experiment to measure g
	Checking for understanding / Homework / Q & A / Reflection etc.
	Take home exercise: feel centripetal force yourself; Quiz next week on newton's laws and projectile motion

Vocabulary



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- Circular 圆形的/圆周的
- Centripetal 向心
- Period 周期
- Frequency 频率
- Rotate/rotation 旋转
- Revolution/revolve 围绕
- Arc 弧长
- Approximate 大约
- Magnitude 大小
- Centrifugal 离心
- Centrifuge 离心机
- Fictitious force 虚拟力

Circular motion



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- Circular Motion is an example of the motion of a body in two dimensions. It is a very specific type of motion because it requires a constant application of force to keep the body on the circular path. This force is called the Centripetal force

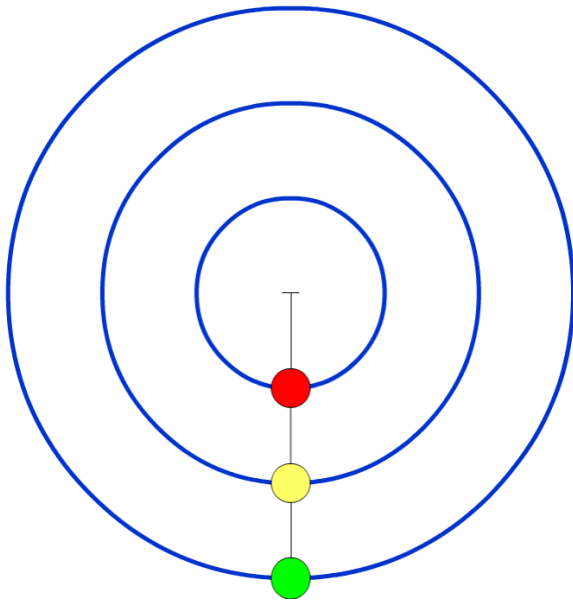


- Perfect circular motions are rare in reality
- Many motions can be approximated to circular motion as they follow a path that is approximately a circle, or part of a circle.
- Any other example you can think of?

Describe uniform circular motion



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- Discussion: the three balls are all doing circular motion. Which ball travels faster?
- *They complete one round of the circle in the same time*
- *Speed of each ball (seems) constant. Direction of speed changes all the time.*
- *The length of path they go through in the same time are not the same*

Describe circular motion – period and frequency



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Period and frequency

- The period T is the time for one complete revolution.

$T = \text{total time} / \text{number of revolutions completed}$

- The frequency f is defined as how many cycles (oscillations, repetitions, revolutions) occur each second.

$f = \text{number of revolutions completed} / \text{total time}$

- The SI unit for frequency is hertz (Hz). Use second as the unit for total time in order to get frequency in the unit of Hz
- One Hertz means one cycle per second. $1 \text{ Hz} = 1 \text{ (cycle) s}^{-1}$

Question: what is the relation between period T and frequency f ?

$$f = \frac{1}{T} \quad \text{or} \quad T = \frac{1}{f}$$

relation between T and f

Describe circular motion – period and frequency



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Quick exercise:

1. use frequency and period to describe the following motions

- a. Earth's rotation
- b. Earth's revolution
- c. The tip of the hour hand/minute hand/second hand of the clock in the classroom

2. The smallest hummingbird's wing-flapping rate is about 80 Hz. How much time is needed for each flap?

Describe circular motion



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- In linear motions, how fast something travels can be described by velocity or speed. What about circular motion?
- How do we judge how fast an athlete runs along a circular track?

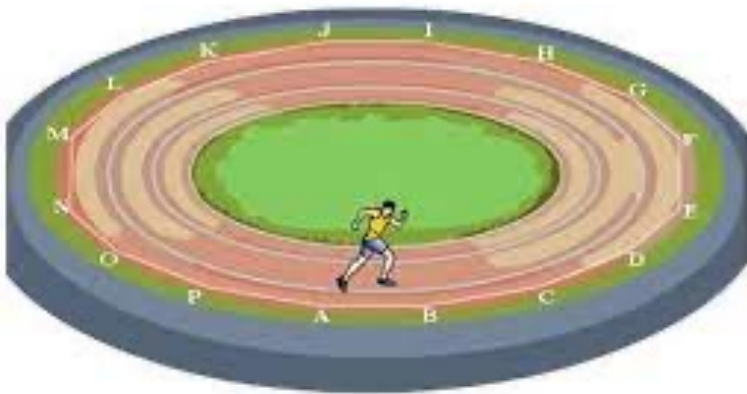
- Linear velocity v :

For uniform circular motion

$$v = \frac{\Delta s}{\Delta t} = \frac{2\pi r}{T} = 2\pi r \cdot f$$

Where Δs is the arc length travelled
 r is the radius of the circular path

- Direction of velocity: along the tangent line of the circular path (changes every moment)



Describe uniform circular motion



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- The linear velocity of minute arms for a big/small clock are obvious not the same. What physical quantity do we use to present they are moving “at the same rate”?
 - angular velocity



For uniform circular motion

$$\omega = \frac{\text{angle swept}}{\text{time taken}} = \frac{\Delta\theta}{\Delta t}$$

Where ω (omega) is the angular velocity, unit is $\text{rad} \cdot \text{s}^{-1}$, or s^{-1}

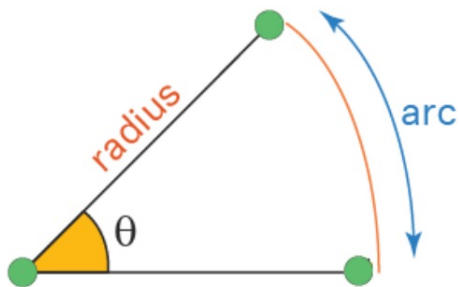
For a complete revolution, $\Delta\theta = 2\pi$ and $\Delta t = T$, so we also have

$$\omega = \frac{2\pi}{T} = 2\pi f$$



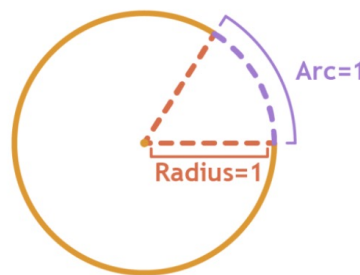
Introducing the SI unit of angle - Radian

- The radian, denoted by the symbol **rad**, is the unit of angle in the International System of Units (SI)



$$\theta = \frac{\text{arc length}}{\text{radius}}$$

1 RADIAN
is when
Arc length = Radius length



Length of arc = $r \cdot \theta$

1 circle = 360 degrees = 2π radians

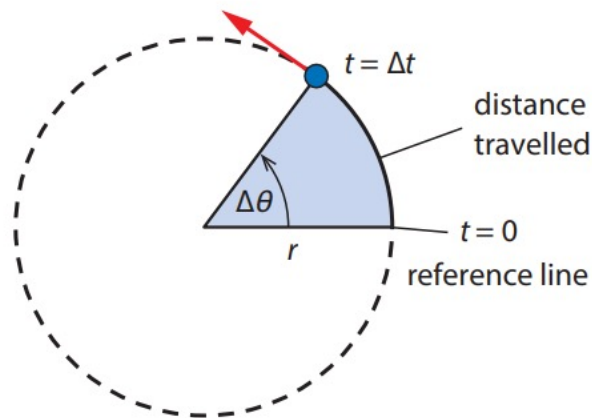
$2\pi \text{ rad} = 360^\circ$

$1 \text{ rad} = \left(\frac{180}{\pi}\right)^\circ \approx 57.3^\circ \text{ or } 57.3 \text{ degrees}$

$1^\circ = \left(\frac{\pi}{180}\right) \text{ rad}$



Describe uniform circular motion



- In a short time Δt the body travels a distance $v\Delta t$ along the circle
- In that time period the angle swept is $\Delta\theta$. If we use rad as the unit for θ , the length of the arc is $r \Delta\theta$

- $v \cdot \Delta t = r \cdot \Delta\theta \rightarrow v = r \cdot \frac{\Delta\theta}{\Delta t} = r \cdot \omega$

Relation between angular velocity and linear velocity:

$$v = \omega r$$

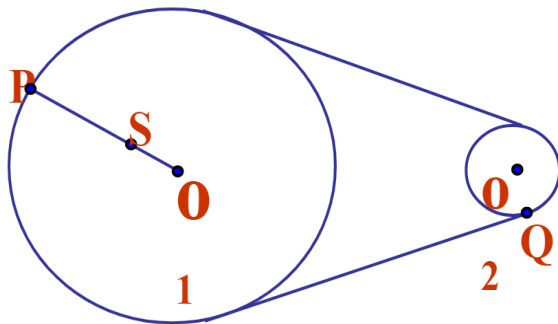
Angular velocity and linear velocity - example



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Two running wheels are connected by a belt. There is no relative motion between the belt and any of the wheels. For the big wheel, $SO:PS = 1:2$.

The radius of the small wheel is $1/3$ of the big wheel



$$v_p : v_Q : v_S = ______ : ______ : ______$$

$$\omega_p : \omega_Q : \omega_S = ______ : ______ : ______$$

$$v_p : v_Q : v_S = 3 : 3 : 1$$

$$\omega_p : \omega_Q : \omega_S = 1 : 3 : 1$$

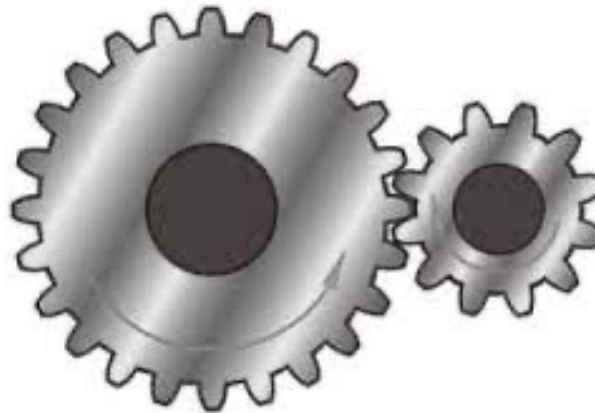
Describe uniform circular motion



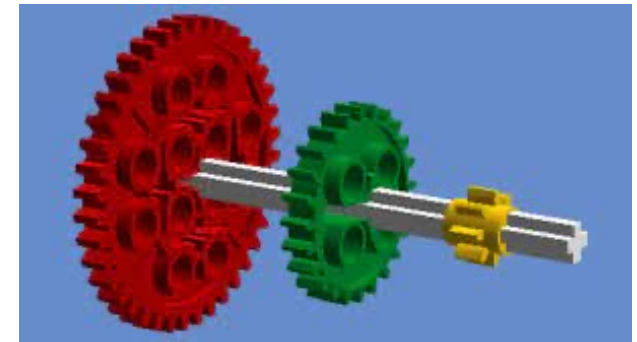
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Same linear velocity



Same linear velocity



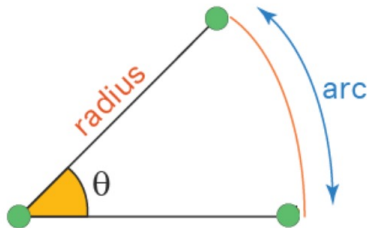
Same angular velocity

Quick quiz



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- Can an object accelerate if it's moving with constant speed?
- What is the period for the earth to orbit the sun?
- A car engine works at 1,800 rpm (revolutions per minute). What is the frequency?
- If I know the period of a circular motion is 2s, what is the angular velocity?
- (arc / radius) in the following picture is 0.9. what is the angle θ in the unit of rad and in the unit of degree?

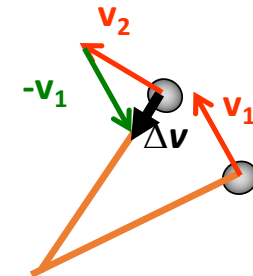
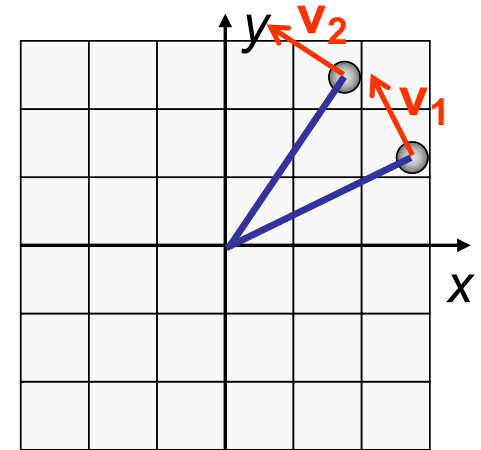


Describe uniform circular motion



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- For circular motion, direction of velocity changes all the time
→ acceleration is not zero
- To find the direction of the acceleration ($a = \frac{\Delta v}{\Delta t}$) we observe two nearby snapshots of the particle
- Now imaging that Δt is very small, or, θ is very small
→ **acceleration points to the center of the circle**
- Alternatively, try to swing a string that's attached to a ball. When the ball is taking circular motion, where does the force pointing? That's the direction of the acceleration



Describe uniform circular motion



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Position

- The path of the motion is a circle, or part of a circle

Time

- Period: how much time taken for one circle
- Frequency: how many rounds per unit time

Velocity

- Linear velocity – at any point, perpendicular to the radius, or, along the tangent of the circle. Magnitude does not change
- Angular velocity is uniform

Acceleration

- Direction: towards the center of the circle
- Magnitude does not change

Try, Think and Discuss



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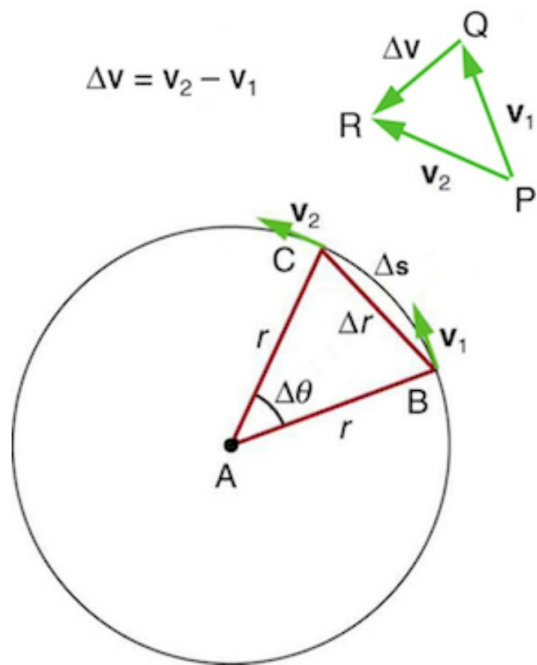
Think about the experiment

- a. Fix the length of the string, when you swing faster and slower, how does the tension change?
- a. Change the length of the string while trying to fix the frequency (e.g. 2Hz, or, two circles per second). How does the tension change?
- ✓ When radius (r) is fixed, tension increases with angular velocity ω or linear velocity v
- ✓ When frequency (f) is fixed (which means angular velocity ω is fixed), tension increases with radius (r)

Centripetal Acceleration



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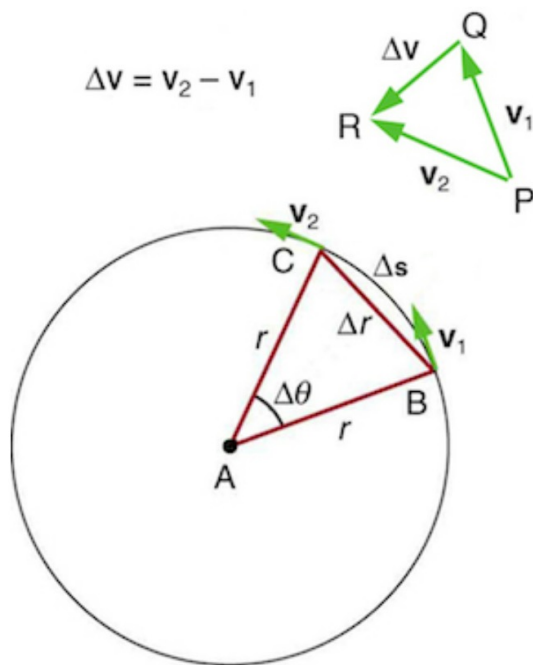
- The directions of the velocity of an object at two different points, B, and C are shown, and the change in velocity Δv , is seen to point roughly toward the center of curvature.
- To see what happens instantaneously, the points B and C must be very close and the $\Delta\theta$ very small.
- We'll then find Δv points directly toward the center of curvature

For uniform circular motions, the acceleration is directed towards the center of the circle, or, centripetal.

Centripetal Acceleration



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- Triangle ABC and triangle PQR are similar triangles (can you figure out the trigonometry?)

$$\frac{\Delta v}{v} = \frac{\Delta r}{r}$$

- When $\Delta\theta$ is very small, $\Delta r = \Delta s$ (length of the arc)

$$\frac{\Delta v}{v} = \frac{\Delta s}{r}$$

- Divide both sides by Δt , and rearrange:

$$\frac{\Delta v}{\Delta t} = \frac{\Delta s}{\Delta t} \cdot \frac{1}{r} \cdot v$$



$$a = \frac{v^2}{r}$$

A body moving along a circle of radius r with speed v experiences centripetal acceleration that has magnitude given by:

$$a = \frac{v^2}{r}$$

as $v = \omega r$,

$$a = \omega^2 r$$

Centripetal Acceleration for uniform circular motions



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$$a = \frac{v^2}{r}$$



As we know $v = \omega r$



$$a = \omega^2 r$$



As we know $v = \frac{2\pi r}{T}$



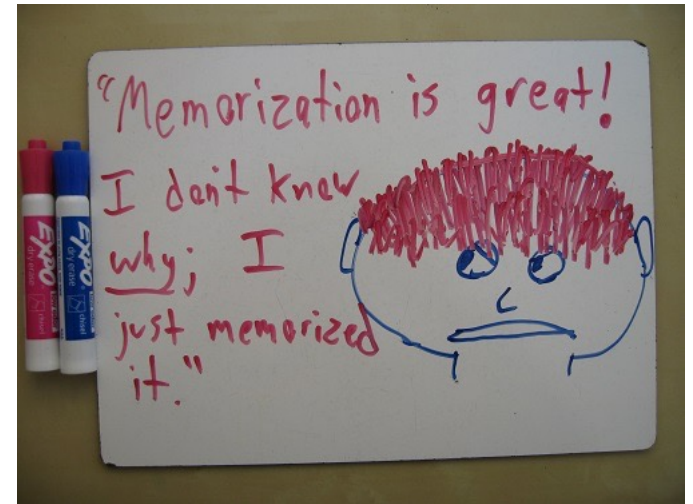
$$a = \frac{4\pi^2 r}{T^2}$$



As we know $f = \frac{1}{T}$



$$a = 4\pi^2 r f^2$$



Choose to memorize them?

Or the easier (smarter) way is to understand the basic concepts of v , ω , T and f

Centripetal Acceleration - example



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Radius of the moon is approximately 1738km. A satellite orbits the moon at an orbit 200 km above the lunar surface. The satellite takes 127 minutes to complete one revolution.

What is the centripetal acceleration of the satellite?

- A. 1.32 m/s^2
- B. 2.23 m/s^2
- C. 3.8 m/s^2
- D. 4.2 m/s^2

Centripetal Acceleration - example



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Centripetal acceleration – man on equator



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The radius of the Earth is $r = 6.4 \times 10^6 \text{m}$. Determine the centripetal acceleration due to the spinning Earth experienced by someone on the equator.

A mass on the equator travels a distance of $2\pi r$ in a time $T = 1 \text{ day}$.

$$\text{Thus: } v = \frac{2 \times \pi \times 6.4 \times 10^6}{24 \times 60 \times 60} = 4.65 \times 10^2 \text{ms}^{-1}$$

$$\text{and so: } a = \frac{(4.65 \times 10^2)^2}{6.4 \times 10^6} = 3.4 \times 10^{-2} \text{ms}^{-2}$$

- How does this acceleration compare to the acceleration due to gravity?
- Is it acceptable to ignore the centripetal acceleration when dealing with an object falling towards earth, and only consider g ?

Centripetal Force



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- If a body moves in a circle, there must be a net force acting on the body, since it is accelerating. If the speed is constant, the direction of the acceleration is towards the center of the circle and therefore that is also the direction of the net force. It is a **centripetal force**.

$$F = ma = m \frac{v^2}{r}$$

or

$$F = ma = m \omega^2 r$$

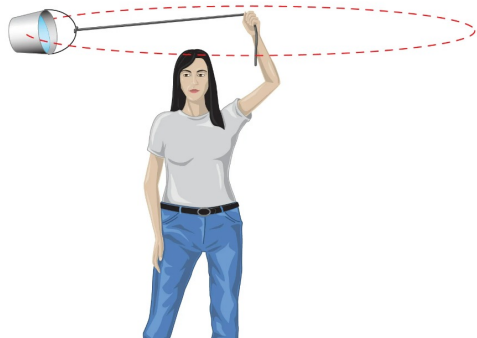
- Centripetal force is not a new type of force. The name specifies the impact of the force on the object. The centripetal force can be friction, gravity, tension, normal force.....etc.

Centripetal Force



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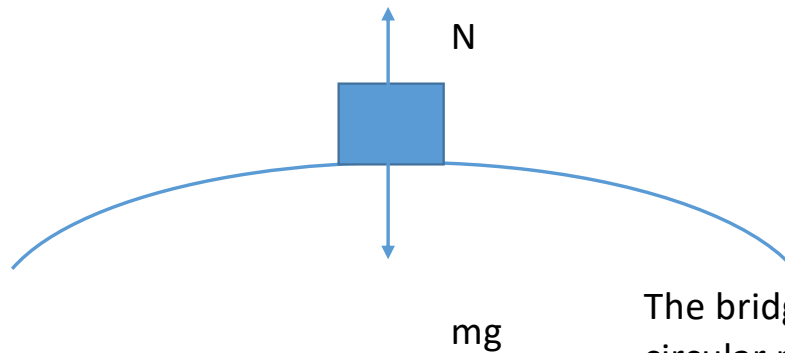
What force (or component of a force) is serving as the centripetal force for the circular motions below?



From Cars on a Bridge to the Space Station



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The bridge can be approximated as part of a circular path with radius R . (R is big)

$$mg - N = m \frac{v^2}{r}$$

$$N = m \left(g - \frac{v^2}{r} \right)$$

As the velocity of the car v goes up, N will be smaller.

- At what velocity will N be 0?
- What happens when N is 0? (recall the lesson on apparent weight)
- How does this explain the weightlessness in space station?

Group activity – design an experiment to measure g



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Available:

- A toy car that you can control the speed.
- A circular track
- A ruler
- Design an experiment to estimate g , acceleration due to gravity

◆ Work in groups, encourage your teammates to explain their ideas in English

◆ Show:

- Measures you need to take/data you need to record down
- A FBD and equation(s) to explain your theory
- Expression of g from the obtained data